Chapter 05

Performance of a Photodiode

- **External Quantum Efficiency**

$$\eta = \frac{Number\ of\ EHP\ generated\ and\ collected}{Number\ of\ incidnet\ photons} = \frac{I_{ph}/e}{P_0/h\nu}$$

- **Sensitivity**

$$S = \frac{Output\ Voltage}{Photocurrent}$$

- **Responsivity**

$$R = \frac{Photocurrent}{Incident\ Optical\ Power} = \frac{I_{ph}}{P_0} \Rightarrow R = \eta\frac{e}{h\nu} = \eta\frac{e\lambda}{hc}$$

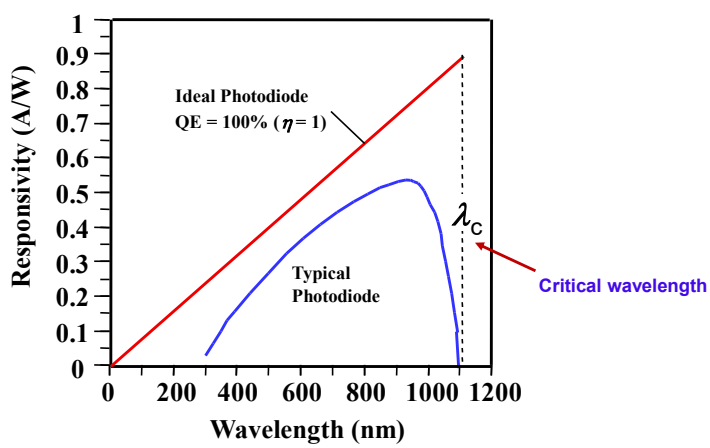
Spectral Responsivity

1

Chapter 05

Ideal and Typical Photodiodes

- **Responsivity vs. wavelength for a photodiode.**



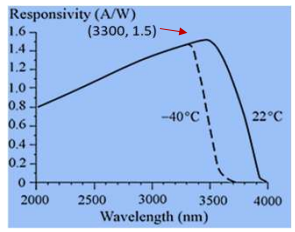
Critical Wavelength - The wavelength at which the responsivity is equal to zero.

2

Chapter 05

Exercise

The following figure shows the responsivity graph of a photodiode manufactured for use in the IR region. It is usually cooled during normal operation.



(a) Determine the quantum efficiency of the photodiode at its peak responsivity at -40°C

(b) Consider the given photodiode with a circular detection area, having a diameter of 0.25 mm at -40°C. Calculate the intensity of light corresponding to a photocurrent of 50 nA at the peak responsivity.

(c) Explain in detail why the cutoff wavelength becomes shorter at lower temperatures

Chapter 05

Exercise

(a) Responsivity $R = \eta \frac{e\lambda}{h\,c}$

$$\eta = \frac{Rhc}{e\lambda} = \frac{1.5 \times (6.63 \times 10^{-34}) \times (3 \times 10^{8})}{(1.602 \times 10^{-19}) \times (3300 \times 10^{-9})}$$

$$= 0.565$$

$$= 56.5\%$$

(b) $R = \frac{I_{ph}}{P_0} \Rightarrow P_0 = \frac{I_{ph}}{R} = \frac{50 \times 10^{-9}}{1.5}$

$$= 33.3 \times 10^{-9} W$$

$$Intensity = \frac{P_0}{A} = \frac{33.3 \times 10^{-9}}{\pi \left(\frac{0.25 \times 10^{-3}}{2}\right)^2}$$

$$= 0.66\ Wm^{-2}$$

(C) The band-gap energy of semiconductors tends to decrease with increasing temperature. The relationship between band gap energy and temperature can be described by the following expression

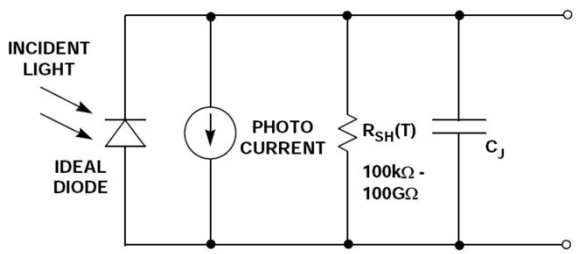
$$E_g(T) = E_g(0) - \frac{\alpha T^2}{T + \beta},$$

where $E_g(0)$, α and β are material constants.

At lower temperatures, the band gap energy increases, which means that photon with lower energies (longer wavelengths) are no longer able to excite electrons across the band gap. As a result, the cut-off wavelength become shorter.

Chapter 05

Photodiode Equivalent Circuit



R_{SH} – Shunt resistance (depend on the temperature)

C_j - junction capacitance of the diode (depend bias voltage)

R_{SH} → Halves every 10^0C temperature rise.

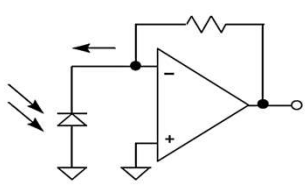
5

Chapter 05

Photodiode Modes of Operation

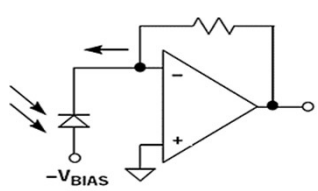
Photodiodes may either be operated with zero bias or reverse bias

- Photovoltaic mode (Zero Bias)
- Photoconductive mode (Reverse Bias)



<u>Photovoltaic</u>

- Zero bias
- No 'Dark' Current
- Linear
- Low Noise (Johnson)
- Precision Applications



<u>Photoconductive</u>

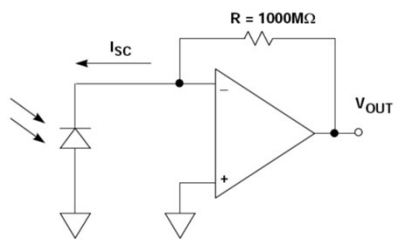
- Reverse bias
- Has 'Dark' current
- Nonlinear
- Higher Noise (Johnson + Shot)
- High Speed Application

6

Chapter 05

Current to Voltage Converter

In general, converting photodiode current to usable voltage is achieved by employing an operational amplifier as a current-to-voltage converter.



Silicon Detector Corp SD-444-11-21-171 Pin Photodiode

- Use an op-amp to reduce the noise.
- Clear night /No moon (0.001fc) → I_{SC}30 pA
- Direct sunlight (1000fc) → I_{SC}30 μA

Calculate the output voltage and sensitivity of the above photodiode under an illumination of 0.001 foot-candles on a clear night.

I_{SC} = 30pA (0.001 fc) → V_{OUT} = 30mV → Sensitivity: 1mV / pA

7

Chapter 05

Leakage Paths and Components

Since the diode current is measured in terms of picoamperes, extreme attention must be given to potential leakage paths in the actual circuit.

Example: -

What is the leakage current between two parallel conductor stripes on a high-quality epoxy-glass PC board, spaced 0.05 inches apart and running parallel for 1 inch. Given that the leakage resistance under these conditions is approximately $1\times10^{11}\Omega$, and a 15-Volt potential difference is applied between these runs.

$$Leakag\ current = \frac{V}{I} = \frac{15\ V}{1\times10^{11}\Omega} = 150\ pA$$

8

Chapter 05

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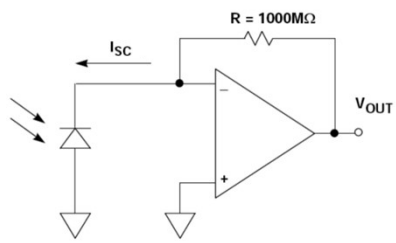

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Chapter 05

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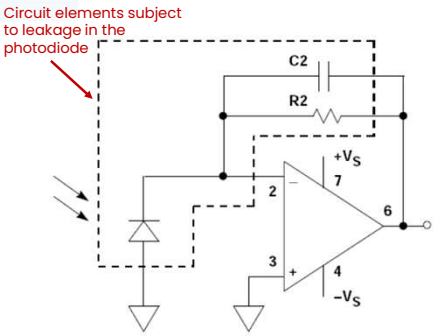
Chapter 05

Leakage Paths and Components



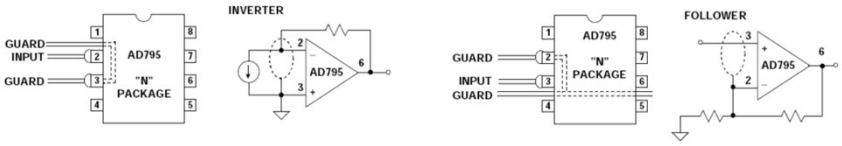
C_2	- Compensation capacitor (Low loss dielectric capacitor)
R_2	– Feedback resistor

- All connection to the summing junction should be kept,
  - Short
  - Clean
  - Free from residues
- If using I/P cable to connect the photodiode to pre-amp
  - Kept it as short as possible.
  - Use Teflon or similar low loss insulator.
- The above points are more obvious towards optimizing accuracy and keeping leakage low. (We use a guard to protect the pin 2)

Chapter 05

Guarding Techniques

- **PCB layout for guarding DIP package**

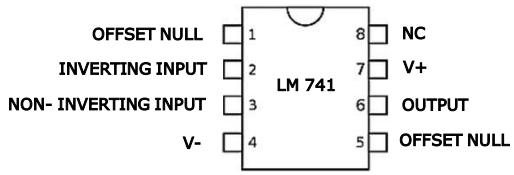


- To reduce the parasitic leakage currents by isolating sensitive inputs from large voltage gradient across PCB.
- Introduce conductive barrier to isolate the input pin.
- In other words guard is a low impedance conductor that completely surrounds I/P.
- The guard voltage is raised to the line's voltage, and it serves to buffer leakage by diverting it away from the sensitive node.

Chapter 05

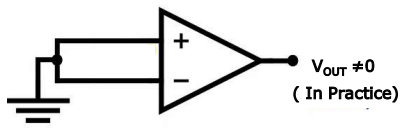
Offset voltage in op-amps (Review)

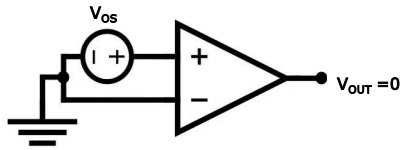
- The offset V_{OS} is the input voltage required to make V_{out} = 0.



❑ **Effect of input offset voltage, V_{OS}**

- The offset V_{OS} is the input voltage required to make V_{out} = 0. It is typically in the range 10μ V – 10 mV



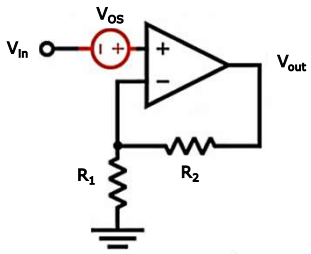


Chapter 05

Offset voltage in op-amps (Review)

- The impact of offset voltage becomes more critical when you are dealing with applications that involve measuring small signals or require high precision

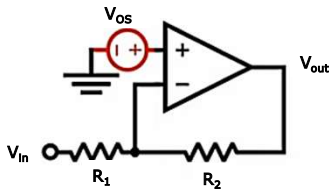
❑ **Non-inverting Amplifier**



$$V_{OUT} = (V_{in} + V_{os})\left(1 + \frac{R_2}{R_1}\right)$$

$$= \left(1 + \frac{R_2}{R_1}\right)V_{in} + \left(1 + \frac{R_2}{R_1}\right)V_{os}$$

❑ **Inverting Amplifier**



$$V_{OUT} = -\left(\frac{R_2}{R_1}\right)V_{in} + \left(1 + \frac{R_2}{R_1}\right)V_{os}$$

* The offset voltage is also amplified → Error

Chapter 05

Offset voltage - TLV9361

TLV9361 ACTIVE

Single, 40-V, 10.6-MHz rail-to-rail output operational amplifier

Product details

Parameters | Package | Pins | Size | Features | Description

Number of channels	1
Total supply voltage (+5 V = 5, ±5 V = 10) (max) (V)	40
Total supply voltage (+5 V = 5, ±5 V = 10) (min) (V)	4.5
Rail-to-rail	In to V-, Out
Slew rate (typ) (V/µs)	25
Vos (offset voltage at 25°C) (max) (mV)	1.7
Iq per channel (typ) (mA)	2.6
Vn at 1 kHz (typ) (nV√Hz)	8.5

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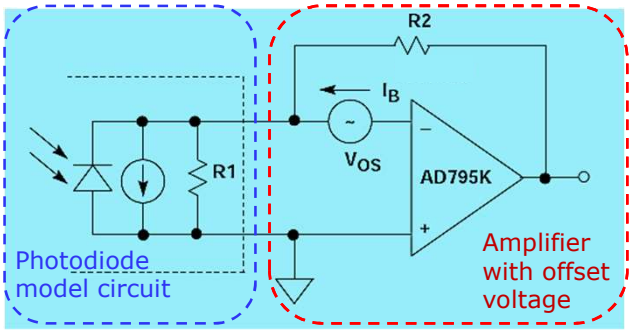
Chapter 05

Offset voltage - photodiode preamp

- An offset voltage and bias current model for the photodiode preamp



- R1 = 1000MΩ @ 25°C (DIODE SHUNT RESISTANCE)
- R1 HALVES EVERY 10°C TEMPERATURE RISE
- I_B DOUBLES EVERY 10°C TEMPERATURE RISE

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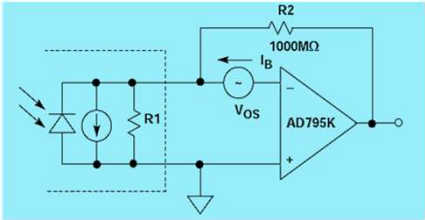
Chapter 05

Exercise

In the following photodiode preamp circuit, the diode shunt resistance (R1) is temperature-dependent, halving with every 10^0C increase. At room temperature (25 ^{0}C), R1 measures 1000 MΩ, and the input bias current is 1 pA. Additionally, it is given that the offset voltage (V_{OS}) for the ADK 795 K BIFET opamp is 0.385 mV at 70 ^{0}C. Calculate the following quantities at 70 ^{0}C

1) DC Noise Gain
2) Error at the output produced by the offset voltage
3) Error of the bias current
4) Error at the output produced by the bias Current
5) Total offset voltage error referred to the output (Total Error RTO)



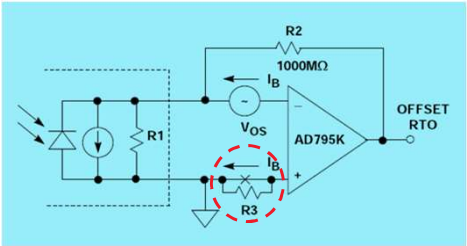
Chapter 05

Output Offset Error

- The primary contributor to the error at high temperature is of course the **bias current** (effect from offset voltage relatively low)

❑ **First order cancellation of this effect**

- An addition of a resistor (R_3) between the noninverting input of the op amp and ground (value of $R_1||R_2$)



- Operating the amplifier at reduced supply voltages, minimizing output drive requirements, and heat sinking.
- The addition of an external offset nulling circuit

Measurements and Instrumentation (BECS 22443)

Lecture 7
Temperature Sensors

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Chapter 06

Temperature Sensors

Common types of temperature transducers

Thermo couple

- Wide range – 184^{0}C to 2300^{0}C
- High accuracy and repeatability
- Needs cold junction compensation
- Low – voltage output

Thermocouple

RTD

- Range: 200^{0}C to 850^{0}C
- Fair linearity
- Requires excitation
- Low cost

RTD

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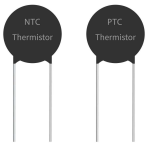
Chapter 06

Temperature Sensors

Thermistor

- Range: 0^0C to 100^0C
- Poor linearity
- Requires excitation
- High sensitivity



NTC PTC

Semiconductor

- Range : -58^0C to 1500^0C
- Linearity :High
- Accuracy : High
- Requires excitation
- 10mV/K, 200mv/K or 14μA/K Typical output

Semiconductor

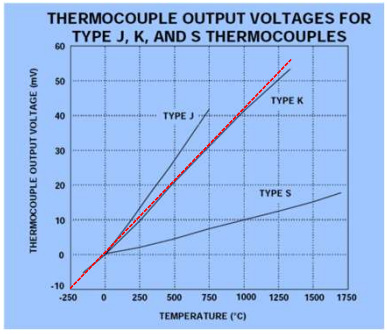
Chapter 06

Thermocouple

❑ **Thermocouple Principle**

Thermocouples rely on the fact that two dissimilar metals joined together produce a voltage, when exposed to a temperature gradient.

❑ **Characteristics (Sensitivity)**



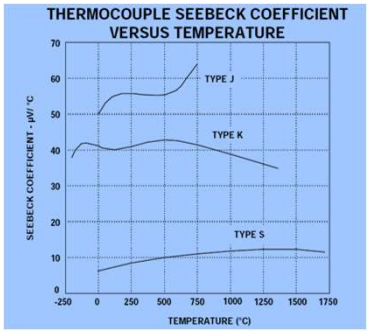
The most common metals used are Iron, Platinum, Rhodium, Rhenium, Tungsten, Copper, Alumel (composed of Nickel and Aluminum), Chromel (composed of Nickel and Chromium) and Constantan (composed of Copper and Nickel).

Chapter 06

Thermocouple

❑ **Characteristics (Seebeck effect)**

An **"emf"** is generated by a temperature difference across the junction of two dissimilar metals. The relationship of Vo to ΔT is called Seebeck coefficient (thermoelectric sensitivity)



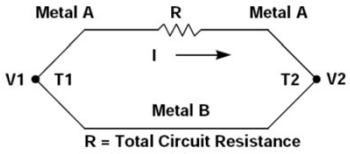
Seebeck coefficient varies with junction temperature

$$\frac{dv}{dt}$$ → First derivative

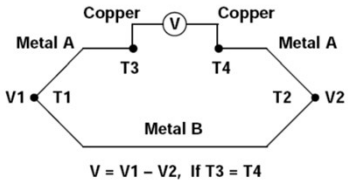
Choose a thermocouple with a stable Seebeck coefficient for accurate temperature measurements across a given range.

Chapter 06

Thermocouple Basics



If we join two dissimilar metals together, creating two similar junctions placed at different locations, there will be a net electromotive force (e.m.f.) in the circuit



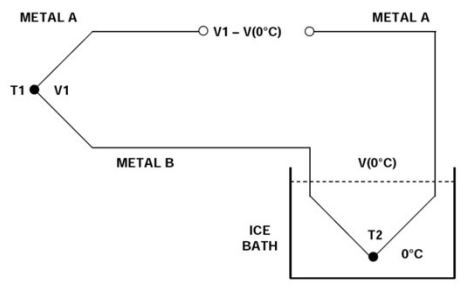
Suppose we break one of the wires, the voltage across the break will be equal to the net thermoelectric e.m.f. of the circuit, and if we measure this voltage, we can use it to calculate the temperature difference between the two junctions

Chapter 06

Cold-junction Compensation

❑ Classical approach

Thermocouple output voltages are frequently specified with a reference junction temperature of 0ºC. To ensure system accuracy, the reference junction must be consistently maintained at a well-defined temperature. Using an ice/water bath provides a relatively straightforward means to establish this reference temperature.



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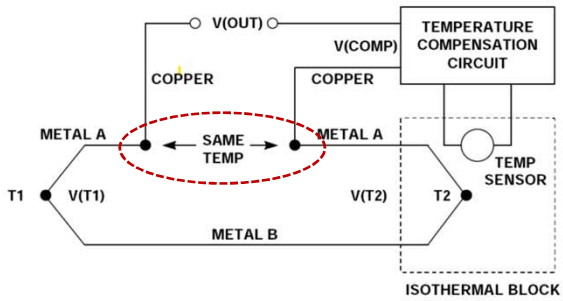
Chapter 06

Cold-junction Compensation - I

❑ Electronical approach

The ice-point reference, with its inconvenient ice/water bath, is generally replaced by electronics. Another type of temperature sensor measures the temperature of the cold junction and is used to inject a voltage into the thermocouple circuit, compensating for the difference between the actual cold junction temperature and its ideal value



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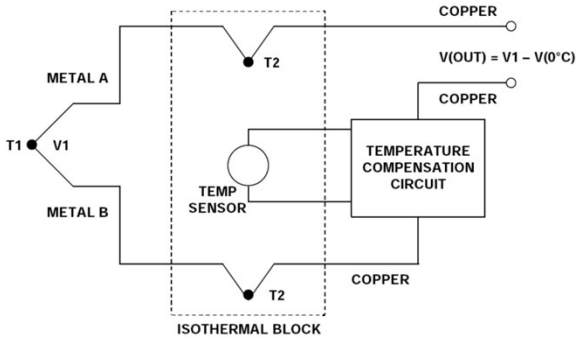
Chapter 06

Cold-junction Compensation - II

❑ **Electronical approach → Junction simplification**

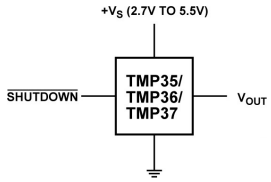
When electronic cold-junction compensation is used, it is common practice to eliminate the additional thermocouple wire and terminate the thermocouple leads in the isothermal block in the arrangement.

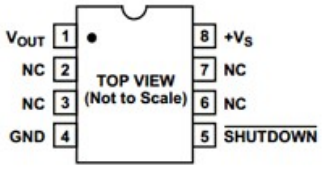


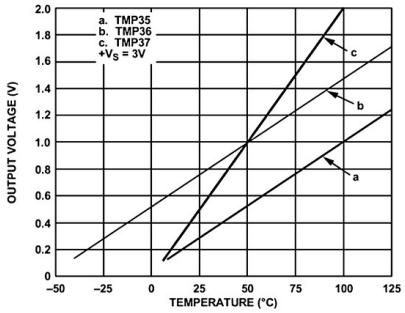
Chapter 06

TMP 35/36/37-SC-Sensor

❑ **Functional block diagram**







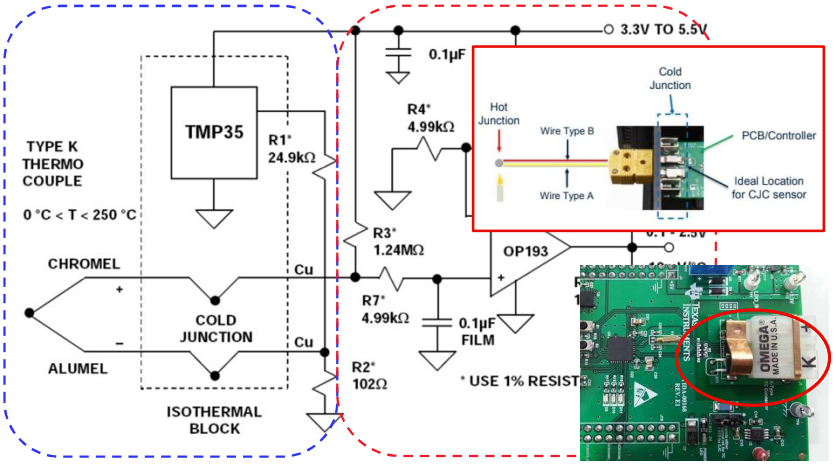
FEATURES

Low voltage operation (2.7 V to 5.5 V)
Calibrated directly in °C
10 mV/°C scale factor (20 mV/°C on TMP37)
±2°C accuracy over temperature (typ)
±0.5°C linearity (typ)
Stable with large capacitive loads
Specified −40°C to +125°C, operation to +150°C

Chapter 06

Amplification Circuit

❑ Type-K Thermocouple



Input stage

Output stage

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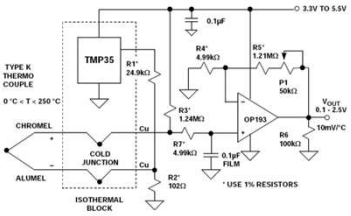
Chapter 06

Amplification Circuit

Seebeck coefficient of thermocouple (Type K) → 41µV/°C
Temperature coefficient of TMP35 sensor → 10 mV/°C



A scaling factor is required to match the temperature coefficients of the TMP35 sensor and the thermocouple, and this is achieved through the resistive voltage divider formed by R1 and R2.

Scaling Factor (Attenuation Factor) ???

$$\text{Attenuation Factor} = \frac{\text{Desired Temperature Coefficient}}{\text{TMP 35 Temperture coefficent}}$$

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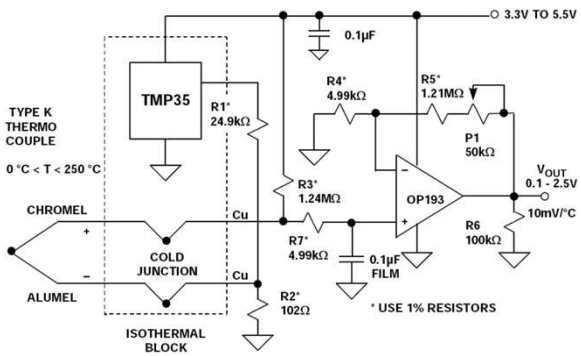
Chapter 06

Exercise

Consider a scenario in which a thermocouple demonstrates a full-scale output voltage of 10.151 mV at a temperature of 250°C within a thermoelectric circuit. Determine the set value for potentiometer P1 required to optimize the operation of the circuit at its full-scale output condition.



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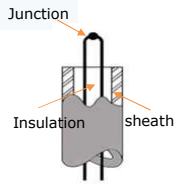
Chapter 06

Measuring Junction Types

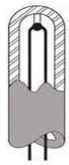


(E) Exposed Junction

Exposed junction thermocouples have their sensing junctions fully exposed to the surrounding environment

Merits → Fast response, Low cost

Demerits → Noise issue, No protection in harsh environment

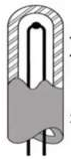


(G) Grounded Junction

Grounded thermocouples have their thermoelectric junctions physically connected to the protective sheath

Merits → High protection from harsh environment

Demerits → Noise issues, Low accuracy



(U) Undergrounded Junction

Ungrounded thermocouples maintain electrical isolation between the junction and the protective sheath

Merits → Free from noise, High protection, High accuracy

Demerits → Slow response

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Chapter 06

Resistance Temperature Detectors

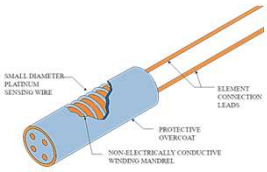
- The resistance temperature detector (RTD) is a sensor whose resistance changes with temperature.

$$R = R_0\,(1 + \alpha_1 T + \alpha_2 T^2 + \alpha_3 T^3 + \ldots + \alpha_n T^n \ldots\ldots)$$

R = Resistance at a given Temperture
R_0 = Resistance at temperature; T = 0K
$\alpha_1, \alpha_2, \alpha_3 \ldots. \alpha_n$ = Constants

❑ Construction

- Typically made with metal elements or metal alloys which have positive temperature coefficient
- A Platinum (Pt) wire has wrapped around a protective coating (RTD element)



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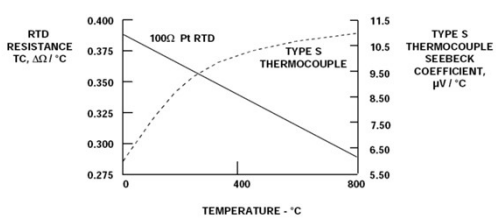
Chapter 06

Resistance Temperature Detectors

❑ Thermocouple Vs RTD's



- Platinum (Pt) RTDs are most common
- 100 Ω, 1000 Ω standard values
- Typical TC = 0.385%/ ^{0}C
- Good linearity (better than thermocouple)

- Required current excitation to produce an output voltage.
- Low TC produce a low output voltage, hence require a high-performance signal conditioning.
- Do not require cold – junction compensation.
- Excessive excitation current can cause self-heating.
  - Self-heating generate measurement errors.
  - Self-heating errors are kept below 0.5^{0}C.

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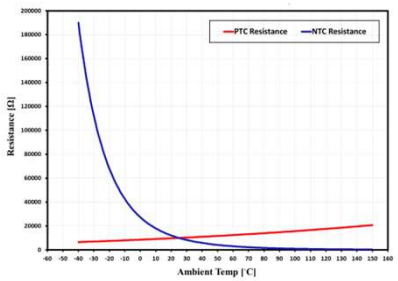
Chapter 06

Thermistors

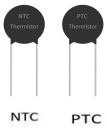
- Thermistors are low-cost, temperature-sensitive resistors constructed of semiconductor materials or ceramic-metal oxide materials, which exhibit positive or negative temperature coefficients, respectively.

- Negative temperature coefficient device are popular than positive temperature coefficient devices.



NTC → T↑ $R_{Therm.}$↓

PTC → T↑ $R_{Therm.}$↑



Chapter 06

Thermistors

❑ Steinhart equation

- The Steinhart equation gives the reciprocal of absolute temperature as a function of the resistance of a thermistor.

$$\frac{1}{T} = A + B\ ln(R) + C\ [ln(R)]^3$$

- R is the resistance of the thermistor at temperature
- T the absolute temperature in Kelvin
- Constants A, B and C can be determined by experimental measurements.

❑ Linearization

- Linearization of the output can be obtained

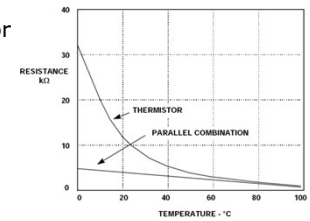
1) Connecting a series metal film resistor

2) Connecting a parallel metal film resistor

Chapter 06

Thermistors

- Paralleling the thermistor, with fixed resistor increases the linearity significantly.
  - Sensitivity
  - Linearity
  - Disadvantage (use only particular range)
  - Value of the resistor



❑ **Calculating the shunt resistor value (R)**

$$R = \frac{R_{T2}\,(R_{T1} + R_{T3}) - 2.R_{T1}.R_{T3}}{R_{T1} + R_{T3} - 2.R_{T2}}$$

- RT_1 → Thermistor resistance at T_1 (lowest temperature in the measurement range)
- RT_3 → Thermistor resistance at T_3 (highest temperature in the measurement range)
- RT_2 → Thermistor resistance at T_2 (midpoint $T_2 = (T_1 + T_2) / 2$)

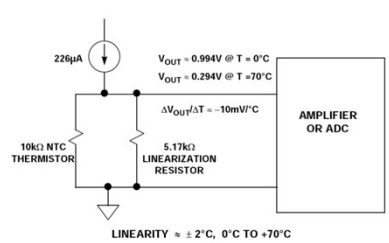
Chapter 06

Thermistors

❑ **Exercise**

For a typical 10 KΩ NTC Thermistor, R_{T1}= 32650 Ω at 0⁰C. R_{T2}= 6532 Ω at 35⁰C and R_{T3}= 1752 Ω at 70⁰C. Determine the value of shunt resistor (R).

$$R = \frac{6532(32650 + 1752) - 2(32650)(1752)}{32650 + 1752 - 2(6532)}$$

$$= 5169.56\ \Omega \approx\ 5.17\ \mathrm{k}\,\Omega$$

- This linearized output can again apply to an ADC for further linearization.